

BE 2nd Year Computer Engineering
Internal Test-2
Data Structure
3CERC3

Max Marks: 20
Time: 70 Mins

****Instructions****

- Answer any 2 questions in each section. All questions carry equal marks

Section A: Attempt Any 2

1. What is a Stack? What are applications of Stack?
2. Can you implement Stack using array? Write the implementation for stack using arrays.
3. Write the implementation for linked list creation?

Section B: Attempt Any 2

4. Calculate the time complexity for following code:

```
int a = 0;
for (i = 0; i < N; i++) {
    for (j = N; j > i; j--) {
        a = a + i + j;
    }
}
```

5. Calculate the time complexity of Merge sort using Masters Theorem for following recurrence

$$T(n) = 2T(n/2) + O(n)$$

6. Find the time complexity for below expression

$$T(n) = 1, \text{ if } n = 1$$

$$T(n) = T(n-1) + 1 \text{ if } n > 1$$

Section C: Attempt Any 2

7. Convert the following as per directed

A) $(A - B/C) * (A/K - L)$ (infix to prefix using stack)

B) $* + AB - CD$ (Prefix to infix using stack)

8. What is a queue explain in detail.

9. What operations queue contains implement all of them?

Section D: Attempt Any 2

10. Implement circular queue.

11. Explain tower of Hanoi with proper implementation.

12. Compare linear queue, circular queue, priority queue and Deque.